Filler-Augmented Training Extends Depth and Prevents State Divergence in Fast-Weight Recurrent Sequence Models

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Abstract

Fast-weight recurrent sequence models with content-addressed memory can compose multi-hop associative retrievals inside a single forward pass, but training at moderate depth reveals three coupled failures: *query-conditioned formation*, in which a fraction of seeds converge to a low-rank readout that collapses when any filler separates the last fact from the query; *state divergence*, in which the residual-stream norm grows exponentially past a sequence-length threshold and the readout collapses onto whichever token direction the explosion aligns with; and a *baseline depth wall* beyond which no healthy seed forms at all. We show that a single training-time modification addresses all three failures simultaneously: insert a uniform-random count of filler tokens between the last fact and the query, with a warm-up-and-ramp schedule and a two-task joint loss. Across trained hop counts $K_{\text{hops}} \in \{3, 4, 5, 6\}$, the recipe drops the mean query-conditioning diagnostic from above the state-persistent boundary to below it, eliminates seeds whose residual-stream norm exceeds 10^3 at $N = 2000$ filler positions, and restores a $4/5$ pass rate at the $K_{\text{hops}} = 6$ cell where the baseline recipe produced $0/3$ healthy seeds. The intervention is scaffold-agnostic: an explicitly two-layer variant of the same architecture solves the $K_{\text{hops}} = 3$ task under the same recipe with target accuracy 0.992 across three seeds. We report a known nuance: at depth the recipe biases the model toward the filler-present distribution, so pass-seeds increasingly need filler at evaluation to reach their headline accuracy. We close by placing the recipe inside the broader landscape of training-time interventions for recurrent recall.

1 Introduction

Fast-weight recurrent sequence models maintain an outer-product associative memory as their hidden state, updating it in place per token. The family runs from Schlag et al. (2021), which cast linear transformers as fast-weight programmers, through the modern DeltaNet lineage (Yang et al., 2024b,a) and its self-referential branch (Irie et al., 2022). On single-hop associative recall, the family closes much of the recall-capacity gap between Mamba-style state-space models (Gu and Dao, 2024; Dao and Gu, 2024) and softmax attention (Arora et al., 2023, 2024). On *multi-hop* composition, where the answer to a query depends on chaining several stored (k, v) pairs, fast-weight models matter for a further reason: the fast-weight state, unlike Mamba's diagonal SSM state, admits a write that composes state-dependent addresses inside a single forward pass. A recent extension of the DeltaNet lineage supplies exactly such a write and composes hop-chains of length $K_{\text{hops}} \leq 5$ inside a fixed three-layer stack at the trained per-hop bank capacity.

At $K_{\text{hops}} = 6$, the baseline training recipe fails categorically: across three replicated seeds, one lands near chance, one appears to converge but is measured to hold the answer in a representation that collapses under a single filler token, and one diverges numerically. The failure is not "the trained network is not accurate enough"; it is three distinct pathologies, all present, all mixed within the seed distribution. We call them *query-conditioned formation*, *state divergence*, and the *depth wall*. They admit separate mechanistic descriptions (Sec. 2.3) but they respond, jointly, to one training recipe: inserting a uniform-random number of filler tokens between the last stored fact and the query, ramped in over a warm-up window.

Contributions. This paper isolates and reports that recipe. Specifically we contribute:

1. A formal description of the filler-augmented training procedure (schedule, loss structure, and hyperparameter defaults) that is architecture-generic within the fast-weight family (Sec. 3).
2. Evidence that the recipe restores a healthy pass rate at the $K_{\text{hops}} = 6$ depth cell where the baseline recipe produces zero healthy seeds (Sec. 5.1).
3. Evidence that the same recipe, at the same hyperparameters, drops the query-conditioning diagnostic below the state-persistent boundary across $K_{\text{hops}} \in \{3, 4, 5, 6\}$ (Sec. 5.2) and eliminates state divergence across the same range (Sec. 5.3).
4. Two ablations: (a) the recipe is scaffold-agnostic, in that an explicitly two-layer variant of the architecture solves under the same procedure with mean target accuracy 0.992 across three seeds (Sec. 6); (b) an occlusion-invariance auxiliary loss steers formation into the same mode, providing a second training-time route to the same effect (Sec. 6).
5. A characterisation of the recipe's principal cost: at $K_{\text{hops}} \geq 4$ the recipe biases the model toward the filler-present distribution, so pass-seeds solve strictly better with filler than without (Sec. 6.3).

Position. The recipe belongs to the catalogue of training-time interventions that shape which representation gradient descent commits to when several representations solve the training distribution equally well (Wang et al., 2024; Olsson et al., 2022; Liu et al., 2023). The distinctive property here is that a single intervention resolves three empirically separable failure modes at once, at the cost of one hyperparameter, and does so through a mechanism that is mechanistically legible: the intervention forces the readout to operate on the fast-weight state rather than the query's local receptive field, and it forces the write on filler tokens to be near-zero. Both of those consequences fall out of the SEP-write geometry the recipe induces.

2 Background

We assume familiarity with fast-weight recurrent layers; readers who want a survey are referred to Schlag et al. (2021); Yang et al. (2024b); Irie et al. (2022). This section fixes notation, states the retrieval task the paper's experiments run on, and names the three failure modes the recipe is designed to address.

2.1 The mixer

At each position t the layer computes query, key, value, and gate projections $q_t = W_q x_t$, $k_t = W_k x_t$, $v_t = W_v x_t$, $\beta_t = \sigma(W_\beta x_t)$, then a state-dependent effective key $k_t^{\text{eff}} = k_t + M_{t-1} k_t$, updates

$$M_t \;=\; M_{t-1} \;+\; \beta_t \cdot k_t^{\text{eff}}\, v_t^\top, \tag{1}$$

and reads $y_t = M_t^\top q_t$. The state $M_t \in \mathbb{R}^{d_k \times d_v}$ is an outer-product associative memory; the effective-key term $M_{t-1} k_t$ is a state *lookup* of the raw key, retrieving the stored key whose associated value most resembles the current input's key, so the write address is a linear function of what is already stored under k_t at the moment of the write.[1] The layer is placed inside a standard Mamba-style scaffold (Gu and Dao, 2024; Dao and Gu, 2024): input projection, grouped convolution, SiLU, z-branch gate, output projection. Throughout the paper we use $L = 3$ layers, $d_{\text{model}} = 256$, $d_k = d_v = 32$, and bf16 autocast unless otherwise stated.

[1] This is the minimal address-side residual in the sense discussed in Irie et al. (2022, 2021). It keeps the recurrence affine in M, which is what allows the layer to be chunk-parallelised at the DeltaNet complexity class (Yang et al., 2024b).

2.2 The retrieval task

The task is K_{hops}-hop factorised associative retrieval, a multi-hop generalisation of the single-hop associative-recall tasks in Arora et al. (2023, 2024) developed for the companion mechanism paper (Voidwolf, 2026). Each row is a concatenation of $K_{\text{hops}} - 1$ intermediate fact banks (each holding K_i (k, v) pairs) and a final bank (holding K_N pairs), separated by SEP tokens, followed by a query token and a target token to be predicted. Bridge assignments between successive banks are drawn as random permutations, so a "confused-entity" traversal cannot land on the correct target by construction. Chance is $1/K_N$; we hold $K_i = 4$ for $i < N$ and $K_N = 8$, giving chance 0.125. This is the "$K_1 = 4/K_N = 8$ matched-difficulty" setting inherited from earlier work on the mechanism; it is the shape at which a positive result cannot be reached by simply outputting some value from a small set.

2.3 Three failure modes at depth

Under the baseline training recipe (no filler, joint half-half loss on a $K_1 = 1$ probe and the K_{hops} target, 15,000 steps, batch 64), the trained model fails at depth in three distinct ways.

(F1) Query-conditioned formation. A fraction of seeds (one third to one half in our replications) converge to a solution whose readout depends on local content adjacent to the query token rather than on the fast-weight state. Inserting even a handful of filler tokens between the last fact and the query collapses the prediction. The failure has a clean training-time diagnostic: QTOK-flip (Voidwolf, 2026), the fraction of the prediction distribution that changes when the query token is half-occluded, above 0.15 at step 5000 predicts the mode at step 15000. Section 5.2 treats this diagnostic in the results.

(F2) State divergence. A separate fraction of seeds (growing with K_{hops}) produce a readout that appears healthy on the training-length distribution but whose residual-stream norm grows exponentially at inference length. On audited baseline seeds, the norm at $N = 500$ inserted filler positions can already reach 10^5; at $N = 2000$ it can exceed 10^{12} (measured on rig-056 $K_{\text{hops}} = 5$ seeds via a post-hoc divergence audit). The readout collapses to whichever vocabulary token happens to align with the direction of the explosion. Different divergent seeds diverge in different directions, so the collapse is seed-specific rather than model-specific.

(F3) The depth wall. Under the baseline recipe, hop counts $K_{\text{hops}} \in \{2, 3, 4, 5\}$ are solved with mean accuracy above 0.94 across three-seed replications. At $K_{\text{hops}} = 6$, three replicated seeds all fail: one lands at chance with mild query-conditioning, one lands at 0.86 target but with QTOK-flip = 0.77 and NaN state on filler probes (a degenerate solve through the query window rather than the state), and one lands at chance with a state that diverges by 10^9-scale norm at $N = 2000$. The failure is not a matter of insufficient budget; each of the three failures reflects a different local optimum the baseline recipe steers seeds into as the mechanism has to compose one more hop than the state's write geometry easily supports.

3 The recipe

The intervention has three components: an insertion, a schedule, and a joint loss.

3.1 Insertion

For each training row containing K_{hops}-hop target structure, before appending the query token, we insert n copies of the SEP token between the last bank's final fact and the query, where

$$n \sim \text{Uniform}\{0, 1, \ldots, n_{\max}(t)\}.$$

One n is drawn per batch (a single filler length applies to all rows within a batch), redrawn every step. $n_{\max}(t)$ is a function of the training step t, described next.

3.2 Schedule

$n_{\max}(t)$ starts at 0, holds at 0 across a warm-up window $[0, T_{\text{warm}}]$, then ramps linearly to a target cap $n^\star$ across $[T_{\text{warm}}, T_{\text{warm}} + T_{\text{ramp}}]$, and holds at $n^\star$ thereafter:

$$n_{\max}(t) \;=\; n^\star \cdot \text{clip}\left(\frac{t - T_{\text{warm}}}{T_{\text{ramp}}},\; 0,\; 1\right).$$

The warm-up matters: the mechanism needs to form on the short-sequence distribution before it can be pulled onto the filler-present distribution. Skipping warm-up produces seeds that never form the mechanism at all. Throughout the paper we use $T_{\text{warm}} = 2000$ steps, $T_{\text{ramp}} = 2000$ steps, and $n^\star = 200$. These defaults have not been tuned per-depth; Sec. 6 shows they carry the $K_{\text{hops}} = 6$ cell without adjustment.

3.3 Loss

Every step draws two batches of equal size: a $K_1 = 1$ probe batch with no filler, and a K_{hops} target batch with filler drawn as above. The total loss is the equal-weight average of the two per-token cross-entropies. The probe batch keeps the write geometry on short sequences aligned with the target task's write geometry; ablating it (target-only loss with filler) produces seeds that solve the filler-present distribution but not the filler-free distribution and vice versa.

3.4 What it changes about the write

The recipe's effect can be read out of Eq. 1: on filler (SEP) positions the gate β_t must not write meaningful content into M, because the target loss punishes any state accumulation that survives up to the query position. This has two consequences. First, the readout $M_T q_T$ at the query position is forced to retrieve from what was stored at the fact positions, not from whatever the last few positions before the query happened to contain. That fixes the query-conditioned failure (F1) by construction. Second, the state stops accumulating on filler tokens, which fixes the state divergence failure (F2) at its geometric cause: the divergence traced by Voidwolf (2026) is precisely a runaway accumulation of non-negligible writes on many SEP positions.

4 Experimental setup

We report five-seed replications for each K_{hops} cell in $\{3, 4, 5, 6\}$, plus a three-seed replication for the two-layer ablation. Training runs 15,000 steps at batch 64 with the AdamW optimiser (weight decay 0.01, $\beta = (0.9, 0.95)$), cosine LR schedule with base LR $3 \cdot 10^{-4}$, 200-step warmup, and floor $3 \cdot 10^{-5}$ (10% of base). Gradients are clipped to unit norm before each step. Autocast is bf16. All experiments run on an AMD Radeon RX 7900 XTX (RDNA3, 24 GB, via ROCm 7.2); a chunked implementation of the mixer (Yang et al., 2024b; Voidwolf, 2026) produces a 4.2x wallclock speedup on the $K_{\text{hops}} = 2$ shape and 6.0x on $K_{\text{hops}} = 3$; loop-parity holds at $K_{\text{hops}} = 5$ to $1.5 \cdot 10^{-6}$ relative error.

Metrics.

- **Target accuracy at N=200 fillers** ($\text{target}_{N=200}$): the primary headline. Evaluated on 2000 held-out rows constructed identically to training but with fixed $n = 200$ filler tokens inserted.
- **Target accuracy at N=0 fillers** ($\text{target}_{N=0}$): how the seed behaves on the filler-free distribution. Difference from $\text{target}_{N=200}$ diagnoses filler-dependence (Sec. 6.3).

- **QTOK-flip**: fraction of predictions that change when the query token is half-occluded, at a fixed filler length $N = 200$. Above 0.15 = query-conditioned; at or below 0.15 = state-persistent (Voidwolf, 2026). Passing seeds under the recipe cluster well below the boundary (typically ≤ 0.03).
- **div_norm@N=2000**: residual-stream norm at position seq_len − 1 when evaluated at $N = 2000$ inserted fillers. Above 10^3 = diverging; below 10^2 = healthy (Voidwolf, 2026).

Verdict rules. A seed *passes* its cell if $\text{target}_{N=200} \geq 0.80$, QTOK-flip ≤ 0.15 (state-persistent), and div_norm@N=2000 $< 10^3$. A cell passes if $\geq 4/5$ seeds pass; a cell is a strong pass if all seeds pass. The rules are pre-registered from the earlier design document; the 0.80 threshold sits well above chance 0.125.

5 Results

The main empirical claim: one recipe, unchanged across four hop counts, restores target accuracy, formation mode, and numerical stability on all four cells. The same recipe, at the same hyperparameters, breaks the $K_{\text{hops}} = 6$ baseline wall.

5.1 Depth wall broken at K_hops=6

Table 1 places the recipe's $K_{\text{hops}} = 6$ result next to the baseline's. The baseline recipe produces zero healthy seeds across three replications: one at chance with mild query-conditioning, one nominally solving the target but with a readout that is entirely query-adjacent (QTOK-flip = 0.77), and one with a residual-stream norm of $1.19 \cdot 10^9$ at $N = 2000$. The filler recipe produces four passing seeds out of five with mean $\text{target}_{N=200} = 0.867$ across the passers, QTOK-flip ≤ 0.02 for all five, and $\text{div\_norm}_{N=2000} \leq 78$ on the passers. The one failing seed under the recipe (s3) sits near chance ($\text{target}_{N=200} = 0.312$) with an elevated div_norm of 2043; formation failure and mild divergence co-occur in the same seed rather than manifesting as distinct events.

Table 1. The $K_{\text{hops}} = 6$ cell. Baseline (rig 129) vs filler recipe (rig 136). n = number of replicated seeds. Passer mean and QTOK-flip mean computed across seeds meeting the verdict rule.

recipe	n	pass rate	$\text{target}_{N=200}$ (mean)	passers mean	QTOK-flip (mean)	div_norm (mean, passers)
baseline	3	0/3	0.450 (spread 0.625)	—	0.323	—
filler	5	4/5	0.756	0.867	0.005	76

5.2 Formation mode across depths

Table 2 reports QTOK-flip means across all four depth cells under the recipe. Every cell sits an order of magnitude below the 0.15 query-conditioned threshold; the mean decreases monotonically with depth. Compared to the baseline, where roughly 1/3 of $K_{\text{hops}} = 5$ seeds land query-conditioned by QTOK-flip (Voidwolf, 2026), the recipe pulls essentially every seed into the state-persistent basin.

Table 2. Formation-mode diagnostic (QTOK-flip mean) across depths under the filler-augmented recipe. Lower is state-persistent; the boundary is at 0.15.

K_{hops}	3	4	5	6
rig	063	132	133	136
seeds	5	5	5	5
QTOK-flip mean	0.033	0.013	0.006	0.004

5.3 Divergence suppressed

Under the baseline recipe at $K_{\text{hops}} = 5$, two of three replicated seeds diverge at $N = 2000$ (div_norm $\gg 10^5$). Under the recipe, all five replicated seeds are non-divergent (div_norm ≤ 103 across the range at $K_{\text{hops}} = 5$; ≤ 78 across the four passers at $K_{\text{hops}} = 6$). One seed at $K_{\text{hops}} = 6$ (s3) has an elevated div_norm of 2043, but this co-occurs with that seed's formation failure rather than presenting as a distinct divergence event.

The reading is that the recipe's SEP-write geometry (forcing β_t near zero on filler positions to satisfy the joint loss) attacks both the formation-mode failure (F1) and the divergence failure (F2) at their shared geometric cause: both are consequences of the write engaging on positions where it should be idle.

5.4 Combined result

Table 3 is the paper's headline result. Across $K_{\text{hops}} \in \{3, 4, 5, 6\}$, the recipe produces mean target$_{N=200} \geq 0.756$ (or, restricted to passing seeds, ≥ 0.867), QTOK-flip mean ≤ 0.033, and eliminates divergence across $n = 20$ trained seeds with the exception of the one formation failure at the deepest cell.

Table 3. Combined recipe results across the trained depth range, $n = 5$ seeds per cell.

K_{hops}	target$_{N=200}$ (mean)	QTOK-flip (mean)	passes	non-divergent	filler-dep gap
3	0.995	0.033	5/5	5/5	~ 0.04
4	0.975	0.013	5/5	5/5	~ 0.32
5	0.917	0.006	5/5	5/5	~ 0.50
6	0.756 (passers 0.867)	0.004	4/5	4/5	~ 0.43

6 Ablations and analysis

Two ablations and one nuance.

6.1 Scaffold agnosticism

The recipe is not tied to the specific three-layer scaffold on which the mechanism was originally characterised. An explicitly two-layer version of the architecture (i.e. $L = 2$) trained under the same recipe on $K_{\text{hops}} = 3$ produces target$_{N=200} = 0.992$ mean across three seeds with QTOK-flip ≤ 0.02 on all three (0.000, 0.016, 0.000). The two-layer variant is exactly the "degenerate" solution that $L = 3$ query-conditioned seeds fall into under the baseline recipe (Voidwolf, 2026); when trained explicitly, and under the filler recipe, it composes cleanly. The recipe therefore does not depend on the three-layer scaffold as a site of refinement; it depends only on the fast-weight write geometry.

6.2 Occlusion-invariance auxiliary loss

An alternative training-time route to the same effect is a λ-weighted KL term that penalises the change in the model's last-position logits when a random 30% of positions between the facts and the query are replaced with SEP:

$$\mathcal{L}_{\text{aux}} = \lambda \, \mathrm{KL}(p(\cdot \mid x) \parallel p(\cdot \mid \mathrm{mask}_{30\%}(x))) .$$

On $K_{\text{hops}} = 3$ with $n = 1$ seed at $\lambda = 0.1$ and $n = 2$ seeds at $\lambda = 1.0$ (plus one earlier $\lambda = 1.0$ run that failed to reach target and is excluded from the summary), the auxiliary produces state-persistent formation with target ≥ 0.98 and QTOK-flip ≤ 0.01. Replication is needed before the auxiliary sits alongside the filler recipe as a validated intervention; we include it here as an indication that the filler recipe is not the only training-time route to the same effect, and that both routes share the property of enforcing invariance to intermediate-position content.

6.3 Filler-dependence at depth

The recipe's principal cost is at $K_{\text{hops}} \geq 4$: mean target$_{N=200}$ is high, but mean target$_{N=0}$ is lower, and the gap grows with depth. On $K_{\text{hops}} = 6$ passers, the mean gap is approximately 0.43; individual passer gaps range up to 0.49. Across all five filler-recipe cells, the widest single-seed gap we observed was 0.61 (rig 133 s_3 at $K_{\text{hops}} = 5$). The recipe biases the model toward the filler-present distribution, and pass-seeds increasingly need filler at eval to reach their headline accuracy. The robust property under this recipe is state-persistence and non-divergence; balanced accuracy under both filler conditions is not robust. Downstream use of the recipe should assume a filler-dependence gap of roughly 0.4 at $K_{\text{hops}} = 6$ (individual passers up to 0.49), and should either evaluate with filler or filter for seeds whose target$_{N=0}$ matches target$_{N=200}$.

7 Related work

Training-time interventions that shape the learned circuit. Wang et al. (2024) show that a Transformer trained on 2-hop composition transitions from a memorised solution to a compositional circuit only via grokking, and that the training mixture of atomic vs composed facts controls whether grokking happens at all. Filler-augmented training is closest in spirit to that observation: a single mixture-level intervention flips which solution gradient descent commits to. The Transformer version acts on in-weights composition; the fast-weight version here acts on in-context composition. Olsson et al. (2022) identify a similar training-time phase change for induction heads.

Filler / pause / think tokens. Goyal et al. (2023) and Pfau et al. (2024) show that inserting learnable or fixed pause tokens between a prompt and its answer can buy the model extra transformer forward passes for computation; Hao et al. (2024) generalises this to continuous CoT vectors. Our filler tokens differ: they are inserted at training time (not eval time) and their purpose is not to grant the model extra computation but to force the write geometry to be idle on filler positions. That is an unrelated mechanism which happens to produce filler tokens as an operational side effect.

Recall-capacity work on fast-weight and Mamba-family layers. Arora et al. (2023, 2024) establish the capacity of single-hop associative recall on fast-weight and SSM architectures. Our results are on the multi-hop generalisation of that task, where the failure surface is qualitatively different: the mechanism must *form* a compositional circuit, not just allocate memory to a set of (k, v) pairs.

Fast-weight lineage. The mixer that carries these experiments is a member of the fast-weight family that runs from Schmidhuber (1992, 1993) through Schlag et al. (2021), Yang et al. (2024b), Yang et al. (2024a), Peng et al. (2025), Sun et al. (2024), Behrouz et al. (2025), and the self-referential branch (Irie et al., 2022, 2021). The specific address-side residual we use, and its multi-hop composition properties, are the subject of a companion paper (Voidwolf, 2026). The filler recipe reported here is expected to be architecture-generic within this family, but we have tested it only on the specific mixer described in Sec. 2; other members of the family may require separate verification.

8 Limitations

The recipe has been tested at one architecture, one task family (injective-connectivity factorised multi-hop retrieval), one sequence-length regime (up to 2000 filler positions at eval), and one hardware setup (AMD Radeon RX 7900 XTX with bf16 autocast). The strongest empirical claims (pass rates and QTOK-flip means) come from $n = 5$ replications per cell, which is enough to distinguish "formation mode is steered categorically" from "formation mode has a small drift" but not enough to characterise the tails. The filler-dependence result reports a mean and range but not a distribution over seeds. The two-layer ablation is $n = 3$; the occlusion-invariance auxiliary ablation is $n = 1$ per λ and should be treated as suggestive rather than validated. The $K_{\text{hops}} > 6$ regime is not tested; there is no architectural reason to expect the recipe to fail at $K_{\text{hops}} = 7$ or higher, but our data cannot rule it out.

9 Conclusion

A single training-time change resolves three empirically distinct failure modes of fast-weight recurrent multi-hop composition at depth: query-conditioned formation, state divergence, and the baseline depth wall. The change: uniform-random filler between the last fact and the query, ramped in over a warm-up window, with a joint loss on a filler-free probe batch and a filler-present target batch. The recipe extends the trainable depth envelope by at least one hop, is scaffold-agnostic, and admits a mechanistic reading in terms of the write geometry it forces on filler tokens. Its principal cost is a filler-dependence gap of roughly 0.4 at $K_{\text{hops}} = 6$ (individual passer gaps up to 0.5) that downstream users should plan around. Further training-time interventions with the same joint property (steering formation and suppressing divergence with one lever) are likely to exist in the fast-weight family; the recipe reported here is a first working example.

References

Simran Arora, Sabri Eyuboglu, Aman Timalsina, Isys Johnson, Michael Poli, James Zou, Atri Rudra, and Christopher Ré. Zoology: Measuring and improving recall in efficient language models, 2023.

Simran Arora, Sabri Eyuboglu, Michael Zhang, Aman Timalsina, Silas Alberti, Dylan Zinsley, James Zou, Atri Rudra, and Christopher Ré. Simple linear attention language models balance the recall-throughput tradeoff, 2024.

Ali Behrouz, Peilin Zhong, and Vahab Mirrokni. Titans: Learning to memorize at test time, 2025.

Tri Dao and Albert Gu. Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality, 2024.

Sachin Goyal, Ziwei Ji, Ankit Singh Rawat, Aditya Krishna Menon, Sanjiv Kumar, and Vaishnavh Nagarajan. Think before you speak: Training language models with pause tokens, 2023.

Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces, 2024.

Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. Training large language models to reason in a continuous latent space, 2024.

Kazuki Irie, Imanol Schlag, Róbert Csórdás, and Jürgen Schmidhuber. Going beyond linear transformers with recurrent fast weight programmers, 2021.

Kazuki Irie, Imanol Schlag, Róbert Csórdás, and Jürgen Schmidhuber. A modern self-referential weight matrix that learns to modify itself. In *International Conference on Machine Learning*, 2022. arXiv:2202.05780.

Ziming Liu, Ouail Kitouni, Niklas Nolte, Eric Michaud, Max Tegmark, and Mike Williams. Towards understanding grokking: An effective theory of representation learning, 2023.

Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, and Others. In-context learning and induction heads, 2022.

Bo Peng, Ruichong Zhang, Daniel Goldstein, Eric Alcaide, Xingjian Du, Teddy Ferdinan, Haowen Hou, Przemysław Kazienko, Kranthi Kiran GV, Jan Kocón, and Others. Rwkv-7 "goose" with expressive dynamic state evolution, 2025.

Jacob Pfau, William Merrill, and Samuel R. Bowman. Let's think dot by dot: Hidden computation in transformer language models, 2024.

Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber. Linear transformers are secretly fast weight programmers. In *International Conference on Machine Learning*, 2021.

Jürgen Schmidhuber. Learning to control fast-weight memories: An alternative to dynamic recurrent networks. *Neural Computation*, 4(1):131–139, 1992.

Jürgen Schmidhuber. Reducing the ratio between learning complexity and number of time varying variables in fully recurrent nets. *International Conference on Artificial Neural Networks*, 1993.

Yu Sun, Xinhao Li, Karan Dalal, Jiarui Xu, Arjun Vikram, Genghan Zhang, Yann Dubois, Xinlei Chen, Xiaolong Wang, Sanmi Koyejo, Tatsunori Hashimoto, and Carlos Guestrin. Learning to (learn at test time): RNNs with expressive hidden states, 2024.

Voidwolf. FENRIR: chunk-parallel address-perturbation fast weights for multi-hop associative composition, 2026. Companion paper (in preparation).

Boshi Wang, Xiang Yue, Yu Su, and Huan Sun. Grokked transformers are implicit reasoners: A mechanistic journey to the edge of generalization, 2024.

Songlin Yang, Jan Kautz, and Ali Hatamizadeh. Gated delta networks: Improving mamba2 with delta rule, 2024a.

Songlin Yang, Bailin Wang, Yu Zhang, and Yoon Kim. Parallelizing linear transformers with the delta rule over sequence length, 2024b.